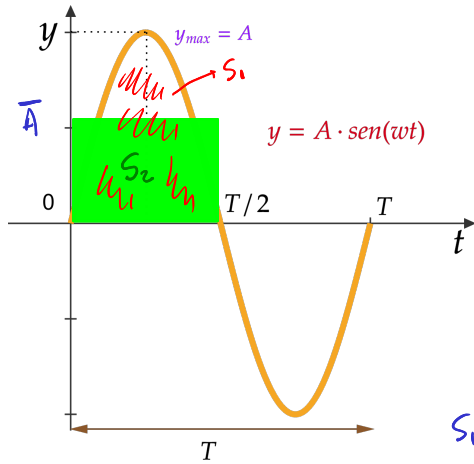


Corriente Alterna

Problema 1



$$S_2 = \bar{A} \cdot T/2$$

||

$$S_1 = \int_0^{T/2} A \cdot \sin(\omega t) dt = \frac{A}{\omega} \cdot [-\cos(\omega t)]_0^{T/2} =$$

$$= \frac{A}{2\pi} \cdot T \cdot \left[-\cos\left(\frac{2\pi}{T} \cdot \frac{T}{2}\right) + \cos\left(\frac{2\pi}{T} \cdot 0\right) \right] =$$

$$= \frac{A}{2\pi} \cdot T \cdot \left[-\cos \pi + \cos 0 \right] = \frac{A}{\pi} T$$

$$\text{Luego } S_1 = S_2 \quad \bar{A} \cdot T/2 = \frac{A \cdot T}{\pi} \Rightarrow \bar{A} = \frac{2 \cdot A}{\pi}$$

Problema 2

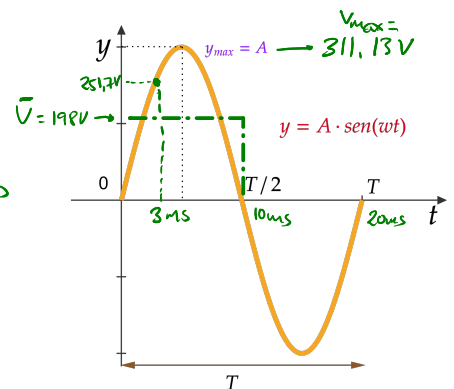
$$V = 220V \text{ y } f = 50\text{Hz} \Rightarrow V_{\text{ef}} = 220V, T = 1/f = 1/50\text{Hz} = 0,02s = 20\text{ms}$$

$$\text{Luego } V_g = \frac{V}{\sqrt{2}} \Rightarrow V_{\max} = V_{\text{ef}} \cdot \sqrt{2} = 311,13V$$

$$\bar{V} = \frac{2}{\pi} V_{\max} = \frac{2 \cdot 311,13V}{\pi} = 198V$$

$$V = V_{\max} \cdot \sin(\omega t)$$

$$V(3\text{ms}) = 311,13V \cdot \sin\left(\underbrace{2 \cdot \pi \cdot 50\text{Hz} \cdot 3\text{ms}}_{\text{¡ojo! En radianes!}}\right) = 251,7V$$



Problema 3

$$V_g = 127 \text{ V} \quad f = 50 \text{ Hz}$$

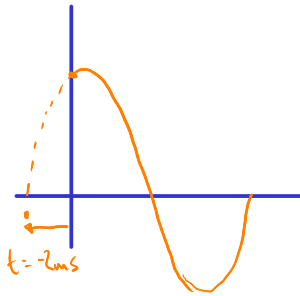
$$V_m = \sqrt{2} \cdot V_g \approx 180 \text{ V}$$

$$V = V_m \cdot \sin(\omega t + \phi) \Rightarrow$$

$$\sin(\omega t + \phi) = 0 \text{ en } t = -2 \text{ ms}$$

$$\omega t + \phi = 0 \text{ en } t = -2 \text{ ms}$$

$$\phi = -\omega t = -2 \cdot \pi \cdot f \cdot (-2 \text{ ms}) = 0,628 \text{ rad}$$



$$V = 180 \cdot \sin(50\pi t + 0,628)$$

Onda adelantada.

Ejemplo: Suma de tensiones desfasadas. $f = 50 \text{ Hz} \rightarrow \omega = 2\pi f = 100\pi \text{ rad/s}$

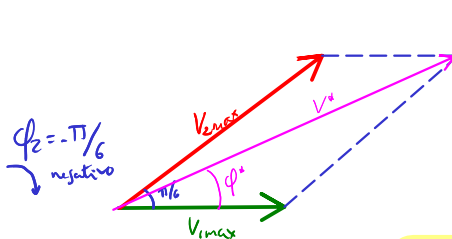
$$V_1 = V_{1\text{max}} \cdot \sin(\omega t) = 180 \cdot \sin(100\pi t) \text{ V}$$

$$V_2 = 311 \cdot \sin(100\pi t + \pi/6) \text{ V} \quad \pi/6 \approx 30^\circ$$

$$V_{1\text{max}} = \sqrt{2} \cdot V_{g1} = \sqrt{2} \cdot 127 \text{ V} = 180 \text{ V}$$

$$V_2 = \sqrt{2} \cdot V_{g2} = \sqrt{2} \cdot 220 \text{ V} = 311,13 \text{ V}$$

Tomarlo negativo. Adelantado.



$$\vec{V}^* = \vec{V}_{1\text{max}} + \vec{V}_{2\text{max}} = V_{1\text{max}} \vec{i} + V_{2\text{max}} \cdot \cos \pi/6 \vec{i} + V_{2\text{max}} \cdot \sin \pi/6 \vec{j}$$

$$\vec{V}^* = (180 + 311 \cdot \cos \pi/6) \vec{i} + 311 \cdot \sin \pi/6 \vec{j} = (449 \vec{i} + 155,5 \vec{j}) \text{ V}$$

$$|\vec{V}^*| = \sqrt{449^2 + 155,5^2} = 475 \text{ V} \quad \text{y} \quad \phi^* = \arctan \frac{155,5}{449} = 0,33 \text{ rad}$$

$$\phi^* = 19,1^\circ$$

$$V = 475 \cdot \sin(100\pi t + 0,33) \text{ V} \quad \left. \begin{array}{l} \phi^* = -0,33 \text{ rad} = -19,1^\circ \end{array} \right\} \text{adelantada.}$$

Problema 4

$$V = 220 \text{ V}, \quad 50 \text{ Hz}$$

$$V_g = \sqrt{2} \cdot 220 \text{ V} = 311 \text{ V}$$

$$\omega = 2\pi f = 100\pi \text{ rad/s}$$

$$I = \frac{V}{R} = \frac{\sqrt{2} \cdot 220 \text{ V}}{50 \Omega} \sin(100\pi t) = 6,22 \cdot \sin(100\pi t) \text{ A}$$

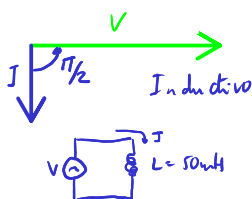
Problema 5

A) $L = 50 \text{ mH}$

$$V_{g1} = 100 \text{ V} \quad f = 50 \text{ Hz}$$

$$V = V_0 \sin(\omega t) = \sqrt{2} \cdot 100 \text{ V} \cdot \sin(100\pi t)$$

$$\omega = 2\pi f = 100\pi \text{ rad/s}$$



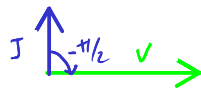
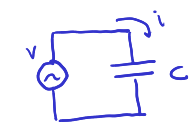
$$X_L = 2\pi f \cdot L = 5000 \cdot \pi \text{ rad/s} \cdot \text{mH} = 5 \cdot \pi \Omega$$

$$I = \frac{V}{R} = \frac{V_0 \sin(\omega t - \pi/2)}{2\pi f \cdot L} = \frac{\sqrt{2} \cdot 100}{5 \cdot \pi} \frac{\text{V}}{\Omega} \sin(100\pi t - \pi/2) = 9 \cdot \sin(100\pi t - \pi/2) \text{ A}$$

Problema 5 B

$$C = 100 \mu\text{F} \quad V = 220\text{V}, 50\text{Hz}$$

$$V = \sqrt{2} \cdot 220\text{V} \cdot \sin(100\pi t) \quad \omega = 2\pi f = 100\pi \text{ rad/s}$$



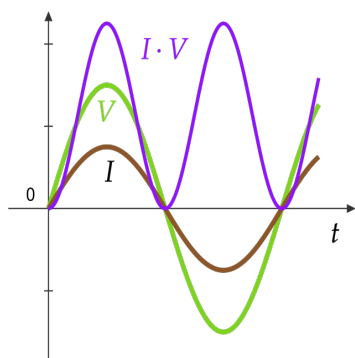
$$I = C \frac{dV}{dt} = I_0 \cdot \sin(\omega t + \pi/2) \text{ adelantada.}$$

$$X_C = \frac{1}{2\pi f \cdot C} = \frac{V_0}{I_0} = \frac{1}{100\pi \cdot 100 \mu\text{F}} = \frac{100}{\pi} \Omega \Rightarrow I_0 = \frac{V_0}{100/\pi \Omega}$$

$$I = \frac{\sqrt{2} \cdot 220\text{V}}{100/\pi \Omega} \sin(100\pi t + \pi/2) = 9.77 \cdot \sin(100\pi t + \pi/2) \text{ A}$$

Problema 6

La potencia media es la integral de los valores en un uniperíodo dividido entre el tiempo considerado. ($T/2$)



$$P_m = \frac{\int_0^{T/2} I \cdot V \cdot dt}{T/2} = \frac{I_0 \cdot V_0 \cdot T/4}{T/2} = \frac{I_0 \cdot V_0}{2}$$

$$\begin{aligned} \int_0^{T/2} I \cdot V \cdot dt &= \int_0^{T/2} I_0 \cdot V_0 \cdot \sin^2(\omega t) \cdot dt = \int_0^{T/2} I_0 \cdot V_0 \cdot \frac{1 - \cos(2\omega t)}{2} \cdot dt = \\ &= I_0 \cdot V_0 \left[\int_0^{T/2} \frac{1}{2} \cdot dt - \frac{1}{2} \int_0^{T/2} \cos(2\omega t) \cdot dt \right] = \\ &= I_0 \cdot V_0 \left[\frac{1}{2} \cdot \frac{T}{2} - \frac{1}{2} \cdot 0 \right] = I_0 \cdot V_0 \cdot \frac{T}{4} \end{aligned}$$

$$\int_0^{T/2} \cos(2\omega t) \cdot dt = \frac{1}{2\omega} \sin(2\omega t) \Big|_0^{T/2} = \frac{1}{2\omega} \cdot \sin(2\omega \cdot \frac{T}{2}) - \sin(2\omega \cdot 0) = 0$$

$$2\omega \cdot \frac{T}{2} = 2 \cdot \frac{2\pi}{T} \cdot \frac{T}{2} = 2\pi \quad \sin(2\pi) = 0$$

$$P_m = \frac{I_0 \cdot V_0}{2} = \frac{I_0}{\sqrt{2}} \cdot \frac{V_0}{\sqrt{2}} = I_{ef} \cdot V_{ef}$$

$$R = 2\Omega$$

$$V = 70 \cdot \sin(100\pi t) \text{ V}$$

$$V_0 = 70 \text{ V}$$

$$I_0 = \frac{V_0}{R} = \frac{70 \text{ V}}{2\Omega} = 35 \text{ A}$$

$$P_{\text{máx}} = I_0 \cdot V_0 = 70 \text{ V} \cdot 35 \text{ A} = 2450 \text{ W}$$

$$P_m = \frac{I_0 \cdot V_0}{2} = 1225 \text{ W}$$

$$P = 2450 \cdot \frac{1 - \cos(200\pi t)}{2} \text{ W}$$

↑ frecuencia doble!

$$\text{O también } \Rightarrow I_{ef} = \frac{I_0}{\sqrt{2}} = \frac{35 \text{ A}}{\sqrt{2}} = 24.75 \text{ A}, \quad V_{ef} = \frac{V_0}{\sqrt{2}} = \frac{70 \text{ V}}{\sqrt{2}} = 49.5 \text{ V} \quad P_m = V_{ef} \cdot I_{ef} = 24.75 \text{ A} \cdot 49.5 \text{ V} = 1225 \text{ W}$$

Problema 7

A) $L = 10 \text{ mH}$ $V_0 = 90 \text{ V}$ $f = 60 \text{ Hz} \Rightarrow \omega = 2\pi f = 120\pi \text{ rad/s}$

$V = 90 \cdot \sin(120\pi \cdot t) \text{ V}$

$V_{ef} = V_0 / \sqrt{2} = 63,64 \text{ V}$

$I_{ef} = I_0 / \sqrt{2}$ $I_{ef} = \frac{V_{ef}}{X_L} = \frac{63,64 \text{ V}}{2\pi f \cdot L} = \frac{63,64 \text{ V}}{120\pi \text{ rad/s} \cdot 10 \cdot 10^{-3} \text{ H}} = 16,88 \text{ A} \Rightarrow I_0 = \sqrt{2} I_{ef} = 23,84 \text{ A}$
 $X_L = 3,77 \Omega$

$P_m = 0$

$Q = I_{ef} \cdot V_{ef} = 16,88 \text{ A} \cdot 63,64 \text{ V} = 1074,3 \text{ VAR}$

$I_{ef}^2 X_L = (16,88 \text{ A})^2 \cdot 3,77 \Omega = 1074,3 \text{ VAR}$

$V_{ef}^2 / X_L = (63,64 \text{ V})^2 / 3,77 \Omega = 1074,3 \text{ VAR}$

$I = 23,84 \text{ A} \cdot \sin(120\pi t - \pi/2)$

B) $C = 30 \mu\text{F}$ $V = 150 \sin(100\pi t)$ $f = 50 \text{ Hz}$ $\omega = 2\pi f$

$X_C = \frac{1}{2\pi f \cdot C} = \frac{1}{100\pi \text{ rad/s} \cdot 30 \cdot 10^{-6} \text{ F}} = 106,1 \Omega$

$I_0 = \frac{V_0}{X_C} = \frac{150 \text{ V}}{106,1 \Omega} = 1,414 \text{ A}$

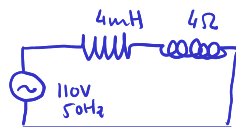
$I = I_0 \sin(100\pi t + \pi/2) = 1,414 \cdot \sin(100\pi t + \pi/2) \text{ A}$

$I_{ef} = \frac{I_0}{\sqrt{2}} = \frac{1,414 \text{ A}}{\sqrt{2}} = 1 \text{ A}$

$V_{ef} = \frac{150 \text{ V}}{\sqrt{2}} = 106,06 \text{ V}$

$Q = I_{ef} \cdot V_{ef} = 106,06 \text{ VAR}$

Problema 8



$V_{ef} = 110 \text{ V}$ $Z = R + j\omega L$

$\omega = 2\pi f = 100\pi \text{ rad/s}$

$R = 4 \Omega$ $\omega \cdot L = 2\pi f \cdot L = 100\pi \text{ rad/s} \cdot 4 \cdot 10^{-3} \text{ H} = 1,26 \Omega$

$|Z| = \sqrt{R^2 + (\omega L)^2} = \sqrt{(4 \Omega)^2 + (1,26 \Omega)^2} = 4,19 \Omega$

$\phi = \arctan\left(\frac{\omega L}{R}\right) = 0,305 \text{ rad}$

$V_0 = \sqrt{2} \cdot V_{ef} = 155,56 \text{ V}$

$I_0 = \frac{V_0}{|Z|} = \frac{155,56 \text{ V}}{4,19 \Omega} = 37,13 \text{ A}$

$I = I_0 \angle -0,305 = 37,13 \text{ A} \angle -17,47^\circ \Rightarrow$

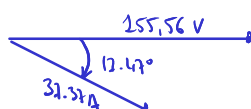
$I = 37,13 \text{ A} \sin(100\pi t - 0,305 \text{ rad})$

$I_{ef} = \frac{I_0}{\sqrt{2}} = 26,42 \text{ A}$

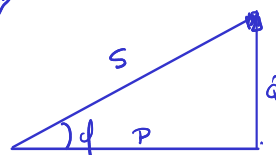
$S = V \cdot I = 110 \text{ V} \cdot 26,42 \text{ A} = 2906,2 \text{ VA}$

$P = S \cdot \cos(\phi) = 2772,07 \text{ W}$

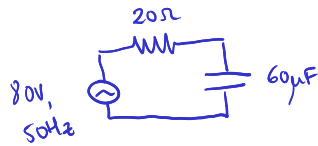
$Q = S \cdot \sin(\phi) = 872,71 \text{ VAR}$



le falta la fase de la intensidad respecto de V. En este caso negativo.



Problema 9



$$1) V = V_0 \angle 0^\circ \quad V_0 = \sqrt{2} \cdot 80V = 113,13V$$

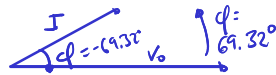
$$2) Z = R - \frac{j}{\omega C} = \frac{20\Omega - \frac{j}{2\pi \cdot 50 \cdot 60 \cdot 10^{-6}}}{100 \cdot \pi \text{ rad/s} \cdot 60 \cdot 10^{-6} \text{ F}} = 20\Omega - \frac{j}{0,019\text{ s}^{-1}} = 20\Omega - 53,05j\Omega$$

$$|Z| = \sqrt{(20\Omega)^2 + (53,05\Omega)^2} = 56,69\Omega$$

$$\phi = \arctan\left(\frac{-1}{\omega \cdot C \cdot R}\right) = -\arctan\left(\frac{1}{\omega \cdot C \cdot R}\right) = -1,21\text{ rad} = -69,32^\circ$$

$$Z = 56,69 \angle -69,32^\circ \Omega$$

El circuito es capacitivo. La intensidad está adelantada



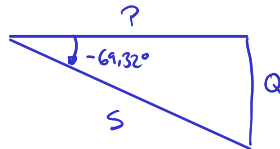
$$I_0 = \frac{V_0}{|Z|} = \frac{113,13V}{56,69\Omega} = 2A$$

$$I = I_0 \angle 69,32^\circ$$

$$I\phi = I_0 / \sqrt{2} = \sqrt{2}A$$

$$I = 2 \cdot \sin(100\pi t + 69,32^\circ) A$$

Potencias



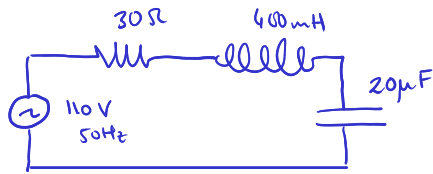
$$S = V_{ef} \cdot I_{ef} = 80V \cdot \sqrt{2}A = 113,13VA$$

$$P = I_{ef}^2 \cdot R = (\sqrt{2}A)^2 \cdot 20\Omega = 40W$$

$$Q = S \cdot \sin(\phi) = 113,13VA \cdot \sin(-69,32^\circ) = -105,85VAR$$

Problema 10

<https://www.translatorscafe.com/unit-converter/es-ES/calculator/series-rlc-impedance/>



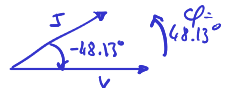
$$V_0 = V_r \cdot 110V = 155,56V \quad \omega = 2\pi f = 100\pi \text{ rad/s}$$

$$Z = R + j\omega L - \frac{j}{\omega C} = 30\Omega + j(100\pi \text{ rad/s} \cdot 400\text{mH}) - \frac{j}{100\pi \text{ rad/s} \cdot 20 \cdot 10^{-6}\text{F}} = 30\Omega + 125,66j\Omega - 159,15j\Omega = 30\Omega - 33,49j\Omega$$

$$|Z| = \sqrt{(30\Omega)^2 + (33,49\Omega)^2} = 44,96\Omega$$

$$\phi = \arctan\left(\frac{-33,49\Omega}{30\Omega}\right) = -0,84 \text{ rad} = -48,13^\circ$$

$$Z = 44,96 \angle -48,13^\circ \Omega$$



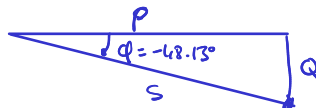
$$I_{ef} = \frac{V_{ef}}{Z} = \frac{110V}{44,96\Omega} = 2,45A, \quad I_0 = \sqrt{2} \cdot 2,45A = 3,46A$$

$$I = I_0 \angle 48,13^\circ$$

$$I = 3,46 \text{ A} \angle (100\pi t + 48,13^\circ) \text{ A}$$

Capacitivo. Adelantada.

Potencias



$$S = V_{ef} \cdot I_{ef} = 110V \cdot 2,45A = 269,5 \text{ VA}$$

$$P = I_{ef}^2 \cdot R = (2,45A)^2 \cdot 30\Omega = 180 \text{ W}$$

$$Q = S \cdot \sin(\phi) = 269,5 \text{ VA} \cdot \sin(-48,13^\circ) = -200,68 \text{ VAR}$$

$\phi = -0,84 \text{ rad}$

$$V_L = I \cdot X_L = 3,46A \cdot \angle 48,13^\circ \cdot \omega L \angle 90^\circ = 3,46A \cdot 100\pi \text{ rad/s} \cdot 400 \cdot 10^{-3}\text{H} \angle 138,13^\circ = 436,8 \angle 138,13^\circ \text{ V}$$

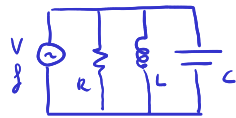
La frecuencia de resonancia cumple que

$$X_L = X_C \quad \omega L = \frac{1}{\omega C} \Rightarrow \omega^2 = \frac{1}{LC} \quad \omega = \frac{1}{\sqrt{LC}}$$

$$\omega = 2\pi f \Rightarrow 2\pi f = \frac{1}{\sqrt{LC}} \quad f = \frac{1}{2\pi \sqrt{LC}}$$

$$f = \frac{1}{2\pi \sqrt{LC}} = \frac{1}{2\pi \sqrt{400\text{mH} \cdot 20\mu\text{F}}} = \frac{1}{2\pi \sqrt{400 \cdot 10^{-3}\text{H} \cdot 20 \cdot 10^{-6}\text{F}}} = 56,269 \text{ Hz}$$

Problema 11



$$X_L = j\omega L$$

$$X_C = \frac{-j}{\omega C}$$

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{j\omega L} - \frac{\omega C}{j} =$$

$$= \frac{(j\omega L)j + Rj - (\omega C)(j\omega L) \cdot R}{R(j\omega L)j}$$

$$\frac{1}{Z} = \frac{-\omega L + Rj - R\omega^2 LCj}{-R\omega L} = \frac{-\omega L + R(1 - \omega^2 LC)j}{-R\omega L} = \frac{1}{R} - \frac{j(1 - \omega^2 LC)}{\omega L}$$

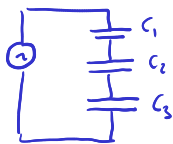
$$Z = \frac{-R\omega L}{-\omega L + R(1 - \omega^2 LC)j} \cdot \frac{-\omega L - R(1 - \omega^2 LC)j}{-\omega L - R(1 - \omega^2 LC)j} = \frac{R\omega^2 L + R^2\omega L(1 - \omega^2 LC)j}{(\omega L)^2 + R^2(1 - \omega^2 LC)^2}$$

$$\phi = \arctan\left(\frac{R^2\omega L(1 - \omega^2 LC)}{R\omega^2 L}\right) = \arctan\left(\frac{R(1 - \omega^2 LC)}{\omega}\right)$$

$$|Z| = \sqrt{\left[\frac{R\omega^2 L}{(\omega L)^2 + R^2(1 - \omega^2 LC)^2}\right]^2 + \left[\frac{R^2\omega L(1 - \omega^2 LC)}{(\omega L)^2 + R^2(1 - \omega^2 LC)^2}\right]^2}$$

$$I = \frac{V_0}{|Z|} \cos(\omega t - \phi)$$

Problema 12



$$Z_1 = \frac{-j}{\omega C_1}$$

$$Z_2 = \frac{-j}{\omega C_2}$$

$$Z_3 = \frac{-j}{\omega C_3}$$

$$Z = Z_1 + Z_2 + Z_3 = \frac{-j}{\omega C_1} - \frac{j}{\omega C_2} - \frac{j}{\omega C_3} = -\frac{j}{\omega} \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

⇒ el circuito se puede sustituir por uno con un condensador equivalente de capacidad

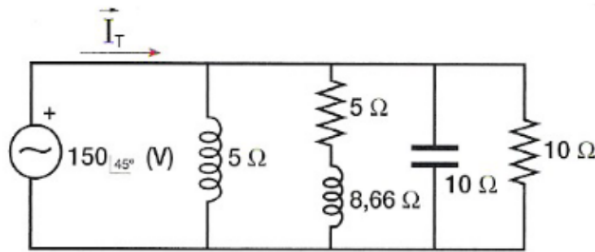


$$Z_{eq} = \frac{-j}{\omega C_{eq}} = -\frac{j}{\omega} \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Si estuvieran en paralelo, la cap. equivalente sería $C = C_1 + C_2 + C_3$

Problema 13 A



$V = 150 \angle 45^\circ \text{ V}$ 150V de tensión máxima cuando me lo dan así.

a) $Z_1 = j\omega L = j5 \Omega$ ¡Cuidado! me dan ya el valor calculado, en impedancia.

b) $Z_2 = 5 \Omega + j8.66 \Omega = 10 \angle 60^\circ \Omega$

c) $Z_3 = -j10 \Omega$ d) $Z_4 = 10 \Omega$

Lo paso a admitancias: $Y_1 = \frac{1}{5 \Omega} = 0.2 \angle -90^\circ \text{ S}$ / $Y_2 = \frac{1}{10 \angle 60^\circ \Omega} = 0.1 \angle -60^\circ \text{ S}$ ($0.1 \cos(-60^\circ) + 0.1 \sin(-60^\circ)j$)

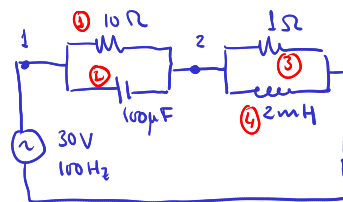
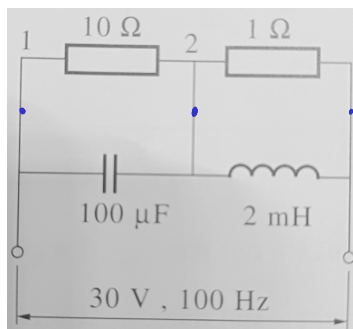
$Y_3 = \frac{1}{-j10 \Omega} = 0.1j \text{ S} = 0.1 \angle 90^\circ \text{ S}$ / $Y_4 = \frac{1}{10 \Omega} = 0.1 \text{ S}$

$Y = Y_1 + Y_2 + Y_3 + Y_4 = [-0.2j + 0.1 \cdot \cos(-60^\circ) + 0.1 \cdot \sin(-60^\circ)j + 0.1j + 0.1] \text{ S} =$

$Y_{eq} = (0.15 - 0.18j) \text{ S} \Rightarrow Z_{eq} = \frac{1}{Y_{eq}} = \frac{1}{0.23 \angle -50.19^\circ \text{ S}} = 4.27 \angle 50.19^\circ \Omega$

$I = \frac{V}{Z} = \frac{150 \angle 45^\circ \text{ V}}{4.27 \angle 50.19^\circ \Omega} = 35.14 \angle -5.19^\circ \text{ A}$

Problema 13 B



$\omega = 2\pi f = 200\pi \text{ rad/s}$

De 1 a 2
 $Z_1 = 10 \Omega$
 $Z_2 = -\frac{j}{\omega C} = -\frac{j}{200\pi \text{ rad/s} \cdot 100 \mu\text{F}} = -15.92j \Omega$

$Y_1 = \frac{1}{10} \text{ S} = 0.1 \text{ S}$

$Y_2 = \frac{1}{-15.92j} \text{ S} = 0.0628j \text{ S}$

$Y_{12} = Y_1 + Y_2 = (0.1 + 0.0628j) \text{ S} = 0.118 \angle 32.13^\circ \text{ S}$

$Z_{12} = \frac{1}{Y_{12}} = \frac{1}{0.118} \angle -32.13^\circ \Omega = 8.468 \angle -32.13^\circ \Omega = (7.17 - 4.5j) \Omega$

De 2 a 3

$Z_3 = 1 \Omega$

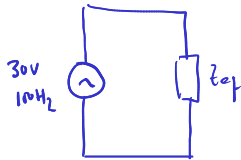
$Z_4 = \omega Lj = 200\pi \text{ rad/s} \cdot 2 \cdot 10^{-3} \text{ H}j = 1.26j \Omega$

$Y_3 = 1 \text{ S}$

$Y_4 = \frac{1}{1.26j} \text{ S} = -0.79j \text{ S}$

$Y_{23} = Y_3 + Y_4 = (1 - 0.79j) \text{ S} \Rightarrow Z_{23} = \frac{1}{Y_{23}} = \frac{1}{1.278 \angle -38.51^\circ \text{ S}} = (0.61 + 0.48j) \Omega = 0.78 \angle 38.51^\circ \Omega$

$Z_{eq} = Z_{12} + Z_{23} = (7.17 - 4.5j) \Omega + (0.61 + 0.48j) \Omega = (7.78 - 4.02j) \Omega = 8.75 \angle -27.32^\circ \Omega$



$$Z_{eq} = 8,75 \angle -27.32^\circ \Omega \quad \phi = -27.32^\circ \text{ atrasada, capacitivo.}$$

$$i = \frac{V_o}{Z} \sin(200\pi t + 27.32^\circ) = \frac{42,43V}{8,75\Omega} \sin(200\pi t + 27.32^\circ) = 4,84 \sin(200\pi t + 27.32^\circ) \text{ A.}$$

$i_{\max}$

$$V_o = \sqrt{2} \cdot V_{ef} = 30V \cdot \sqrt{2} = 42,43V$$

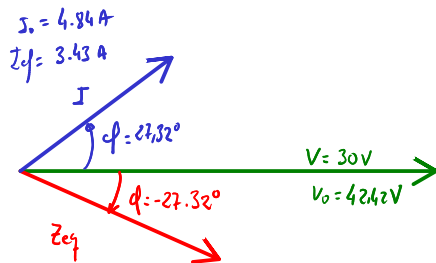
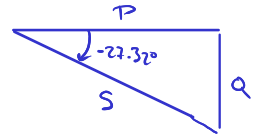
$$I_{ef} = \frac{V_{ef}}{Z} = \frac{30V}{8,75\Omega} = 3,43 \text{ A}$$

Potencia activa.

$$S = V_{ef} \cdot I_{ef} = 30V \cdot 3,43 \text{ A} = 102,6 \text{ VA}$$

$$P = V_{ef} \cdot I_{ef} \cdot \cos(-27.32^\circ) = 91,16 \text{ W}$$

$$Q = V_{ef} \cdot I_{ef} \cdot \sin(-27.32^\circ) = -47,09 \text{ VAR}$$



Como el ángulo de la intensidad en este caso se toma desde V , la intensidad tiene la expresión compleja $I = I_o \angle \phi = 4,84 \text{ A} \angle 27.32^\circ$

$$V_{12_0} = I_o \cdot Z_{12} = (4,84 \text{ A} \cdot \angle 27.32^\circ) \cdot (8,468 \angle -32.13^\circ \Omega) = 40,98V \angle -4,81^\circ = (40,83 - 3,44j)V$$

$$V_{12_0} = V_{12_0} / \sqrt{2} = 28,97V$$

$$V_{23} = I_o \cdot Z_{23} = (4,84 \text{ A} \angle 27.32^\circ) \cdot (0,78 \Omega \angle 38.51^\circ) = 3,77 \angle 65.83^\circ V = (1,545 + 3,44j)V$$

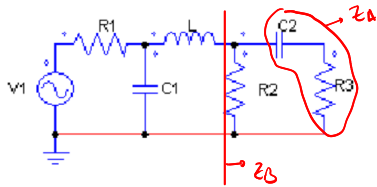
$$V_T = V_{12} + V_{23} = (1,545 + 3,44j)V + (40,83 - 3,44j)V = 42,38 \angle 0.005^\circ V$$

$$\hookrightarrow V_{ef} = \frac{42,38V}{\sqrt{2}} = 29,96V \approx 30V$$

Comprobar que está bien por Kirchhoff.

Problema 13 C

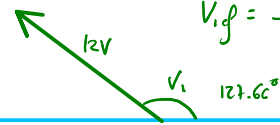
Hallar I (corriente que genera V_1) en el circuito inferior, sabiendo que $R_1 = 0.5\Omega$, $C_1 = 1F$, $L = 1H$, $C_2 = 0.5F$, $R_2 = 3\Omega$, $R_3 = 2\Omega$ y la tensión $V_1 = 12 \cdot \sin(\omega t + 14)$.



$$V_1 = 12V / 14rad = 12V / 2.28rad = 12V / 127.66^\circ$$

$$\omega = 100\pi rad/s \quad f = 50Hz$$

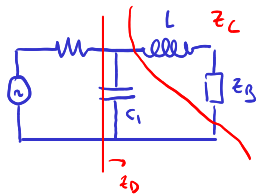
$$V_1 \cdot f = \frac{V_1}{\sqrt{2}} = 8.485V$$



$$Z_A = X_{C2} + R_3 = 2\Omega + \frac{-j}{\omega C_2} = 2\Omega - \frac{2j}{\omega} = 2\Omega - \frac{j}{50\pi} \approx 2\Omega$$

$$Z_B^{-1} = \frac{1}{R_2} + \frac{1}{Z_A} = \frac{1}{3\Omega} + \frac{1}{2\Omega} = \frac{5}{6\Omega}$$

$$Z_B = 6/5 \Omega$$

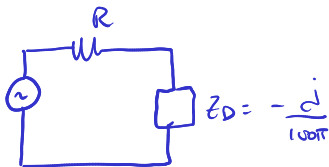


$$Z_C = X_L + Z_B = (100\pi rad/s \cdot 1H) + 6/5\Omega = 314.16 \angle 89.28^\circ \Omega$$

$$Z_D^{-1} = \frac{1}{Z_C} + \frac{1}{X_{C1}} = \frac{1}{314.16 \angle 89.28^\circ} + 100\pi \angle 90^\circ \approx 100\pi \angle 90^\circ$$

$$X_{C1} = \frac{-j}{\omega C_1} = \frac{-j}{100 \cdot \pi \cdot 1F} = \frac{1}{100\pi} \angle -90^\circ \Omega \quad Z_D^{-1} = 100\pi \angle 90^\circ \Omega^{-1} \quad Z_D = \frac{1}{100\pi} \angle -90^\circ \Omega$$

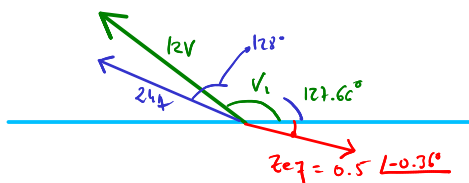
$$Z_D = \frac{-j}{100\pi} \approx X_{C1}$$



$$Z_{eq} = R - j/100\pi = (0.5 - \frac{j}{100\pi}) \Omega = 0.5 \angle -0.36^\circ \Omega$$

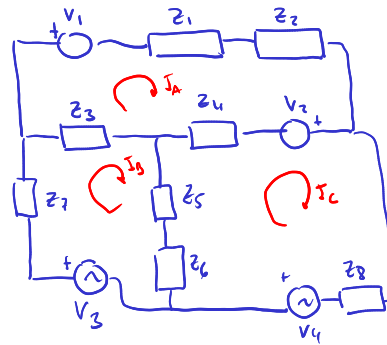
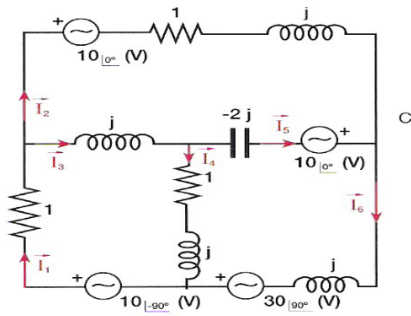
$$I = \frac{V}{Z_{eq}} = \frac{12 \angle 127.66^\circ}{0.5 \angle -0.36^\circ} \frac{V}{\Omega} = 24A \angle 128.0^\circ$$

$$I_g = \frac{24}{\sqrt{2}} = 16.97A$$



Intensidad adelantada
respecto de la tensión.
Circuito Capacitivo.

Problema 13.D



$$\begin{aligned} V_1 &= 10 \angle 0^\circ \text{ V} \\ V_2 &= 10 \angle 0^\circ \text{ V} \\ V_3 &= 10 \angle -90^\circ \text{ V} = -j10 \text{ V} \\ V_4 &= 30 \angle 90^\circ \text{ V} = 30j \text{ V} \\ Z_1 &= 1\Omega \quad Z_2 = j\Omega \\ Z_3 &= j\Omega \quad Z_4 = -2j\Omega \\ Z_5 &= 1\Omega \quad Z_6 = j\Omega \\ Z_7 &= 1\Omega \quad Z_8 = j\Omega \end{aligned}$$

A) $+V_1 + I_A(Z_1 + Z_2) + V_2 + (I_A - I_C)Z_4 + Z_3(I_A - I_B) = 0$
 $10 + I_A(1 + j) + 10 - 2jI_A + 2jI_C + I_A j - I_B j = 0$
 $I_A - I_B j + 2jI_C = -20 \text{ V} \quad | \text{Ten A, ven V.}$

B) $j(I_B - I_A) + (I_B - I_C)(1 + j) - V_3 + I_B \cdot 1\Omega = 0$
 $I_B(2 + j) - I_A j - I_C(1 + j) = V_3 \quad -I_A j + (2 + j)I_B - (1 + j)I_C = -10j \text{ V}$

c) $I_C j - V_4 + (I_C - I_B)(1 + j) + (I_C - I_A)(-2j) - V_2 = 0$
 $2jI_A - (1 + j)I_B + I_C = V_2 + V_4 = (10 + 30j) \text{ V}$

$$\begin{aligned} \text{A)} \quad I_A - jI_B + 2jI_C &= -20 \text{ V} \\ \text{B)} \quad -I_A j + (2 + j)I_B - (1 + j)I_C &= -10j \text{ V} \\ \text{C)} \quad 2jI_A - (1 + j)I_B + I_C &= 10 + 30j \text{ V} \end{aligned}$$

$$\begin{pmatrix} 1 & -j & 2j \\ -j & 2 + j & -(1 + j) \\ 2j & -(1 + j) & 1 \end{pmatrix} \begin{pmatrix} I_A \\ I_B \\ I_C \end{pmatrix} = \begin{pmatrix} -20 \text{ V} \\ -10j \text{ V} \\ 10 + 30j \text{ V} \end{pmatrix}$$

<https://matrix.resish.com/es/cramer.php>

$$\begin{aligned} \times 5 \quad & 5(3 + 2j)I_B - 5(3 + j)I_C = -150j \\ + \times (3 + j) \quad & -(3 + j)(3 + j)I_B + 5(3 + j)I_C = (10 + 70j)(3 + j) \end{aligned}$$

$$(15 + 10j - 8 - 6j)I_B = 30 + 210j + 10j - 70 - 150j$$

$$(7 + 4j)I_B = -40 + 70j$$

$$I_B = \frac{-40 + 70j}{7 + 4j} = 10j \text{ A}$$

A) $(3 + 2j)I_B - (3 + j)I_C = -30j$

B) $-(3 + j)I_B + 5I_C = 10 + 70j$
 $-2j \cdot A + C$

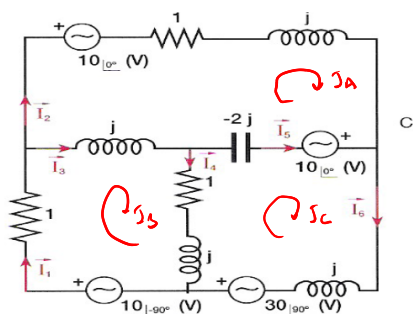
$$I_C = \frac{(10 + 70j) + (3 + j)10j}{5}$$

$$I_C = (2 + 14j) + (6j - 2) =$$

$$I_C = 20j$$

A) $I_A - jI_B + 2jI_C = -20 \text{ V}$

$$I_A = -20 \text{ V} + jI_B - 2jI_C = -20 \text{ V} - 10 \text{ V} + 40 \text{ V} = 10 \text{ A}$$



$$I_A = 10 \text{ A} \quad I_B = 10j \text{ A} \quad I_C = 20j \text{ A}$$

$$I_1 = I_B = 10j \text{ A}$$

$$I_3 = I_B - I_A = (10j - 10) \text{ A} = (-10 + 10j) \text{ A}$$

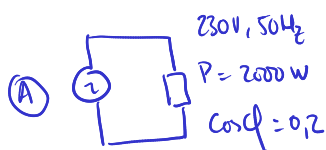
$$I_2 = I_A = 10 \text{ A}$$

$$I_4 = I_B - I_C = 10j - 20j \text{ A} = -10j \text{ A}$$

$$I_5 = I_C - I_A = 20j \text{ A} - 10 \text{ A} = (-10 + 20j) \text{ A}$$

$$I_6 = I_C = 20j \text{ A}$$

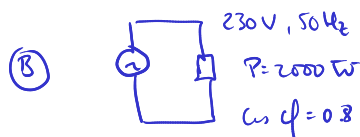
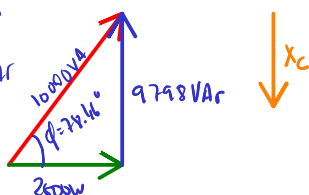
Problema 14



$$\cos \phi = 0.2 \Rightarrow \phi = 78.46^\circ$$

$$S = \frac{P}{\cos \phi} = 10000 \text{ VA}$$

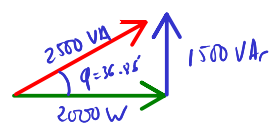
$$Q = S \cdot \sin(\phi) = 9798 \text{ VAR}$$



$$\cos \phi = 0.8 \Rightarrow \phi = 36.86^\circ$$

$$S = \frac{P}{\cos \phi} = \frac{2000 \text{ W}}{0.8} = 2500 \text{ VA}$$

$$Q = S \cdot \sin \phi = 1500 \text{ VAR}$$



Así, en A) le añadimos un condensador, tal que X_C compense la diferencia de VAR.

$$Q = Q_A - Q_B = I_{ef}^2 \cdot |X_C| = \frac{V_{ef}^2}{|X_C|} \Rightarrow |X_C| = \frac{V_{ef}^2}{Q_A - Q_B} = \frac{(230 \text{ V})^2}{9798 \text{ VAR} - 1500 \text{ VAR}} =$$

$$\frac{1}{\omega C} = |X_C| = 6.375 \Omega$$

$$C = \frac{1}{\omega |X_C|} = \frac{1}{2 \cdot \pi \cdot f \cdot |X_C|} = \frac{1}{100 \pi \text{ rad/s} \cdot 6.375 \Omega} = 4.99 \cdot 10^{-4} \text{ F} = 500 \mu\text{F}$$